

Attributes & Key Constraints

Q. Discuss the different types of Attributes in the ER model. Explain the concepts of Super Key, Candidate Key, Primary Key, and Foreign Key with examples.

[DEF] Definition to Remember

An Attribute is a property or characteristic that describes an entity, corresponding to a column in a relational table. A Key is a single attribute or combination of attributes that uniquely identifies a row (tuple) in a table, maintaining data integrity.

1. Types of Attributes

Type	Description	ER Symbol	Example
Simple	Cannot be subdivided further	Single Oval	`Age`, `Gender`
Composite	Can be divided into sub-parts	Single Oval (with child ovals)	`Address` -> Street, City, Zip
Single-Valued	Holds one value per entity	Single Oval	`Date_of_Birth`
Multi-Valued	Holds multiple values per entity	**Double Oval**	`Phone_Number` (mobile + home)
Stored	Physically stored in the DB	Single Oval	`Date_of_Birth`
Derived	Computed from a stored attribute	**Dashed Oval**	`Age` (derived from DOB + current date)

2. Types of Keys

A Key is essential for uniqueness and establishing relationships between tables.

1. Super Key

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2. Candidate Key

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3. Primary Key

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4. Foreign Key

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STUDENT Table	ENROLLMENT Table
Roll_No Name	Roll_No (FK) Course_ID

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Attribute types: Simple, Composite, Single-Valued, Multi-Valued, Stored, Derived.
2. Multi-Valued attribute uses Double Oval; Derived attribute uses Dashed Oval in ER diagrams.
3. Super Key = any set that uniquely identifies a row (may have redundant attributes).

4. Candidate Key = minimal Super Key (no redundant attributes); a table can have multiple.
5. Primary Key = the one Candidate Key selected; cannot be NULL; only one per table.
6. Foreign Key = references Primary Key of another table; enforces Referential Integrity.
7. Super Key Candidate Key Primary Key (each is a subset/specific case of the previous).

[RECALL] Quick Recall

Simple/Composite -> Single/Multi-Valued (Double Oval) -> Stored/Derived (Dashed Oval) -> Super Key -> Candidate Key (minimal) -> Primary Key (chosen, no NULL) -> Foreign Key (referential integrity)